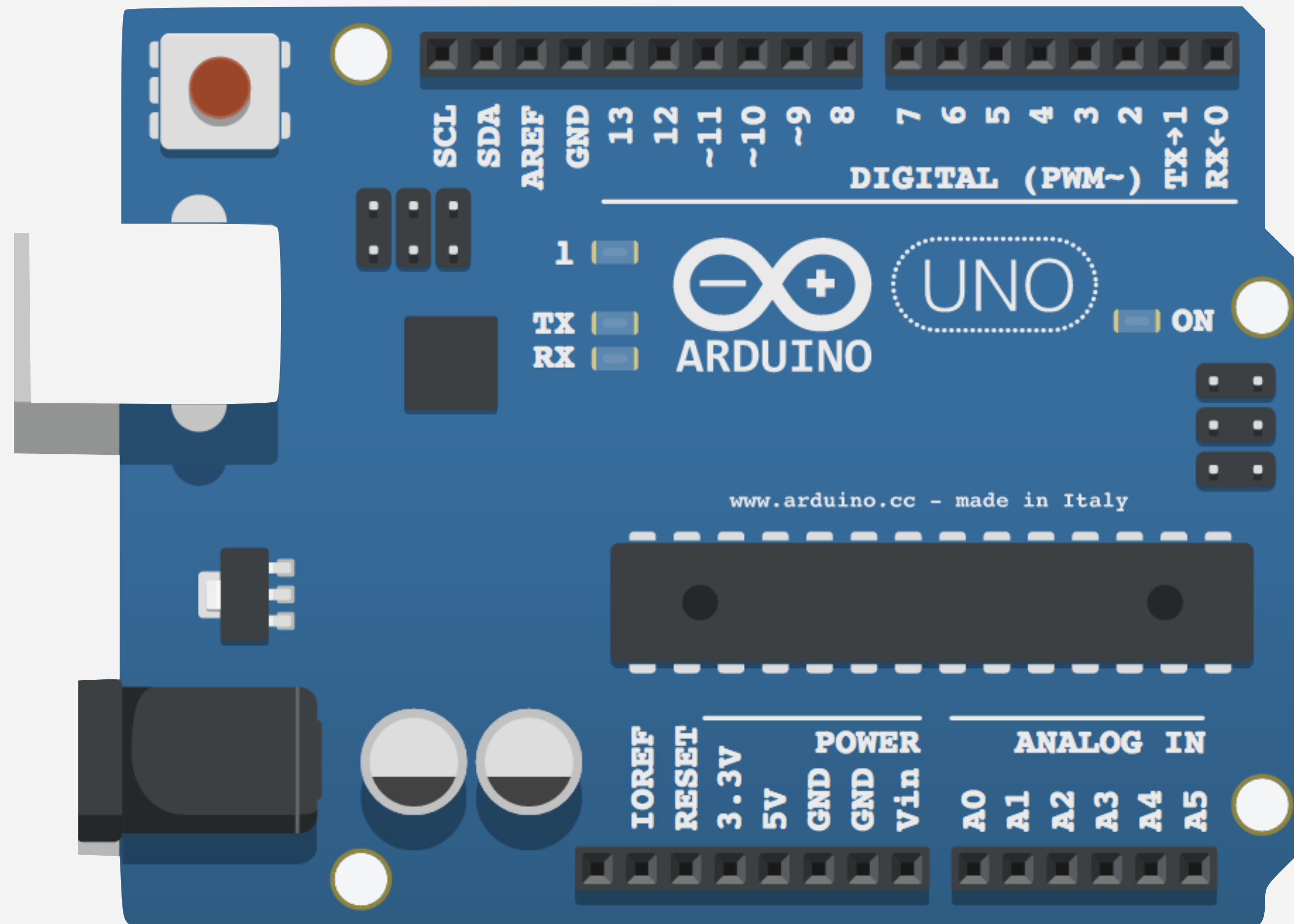
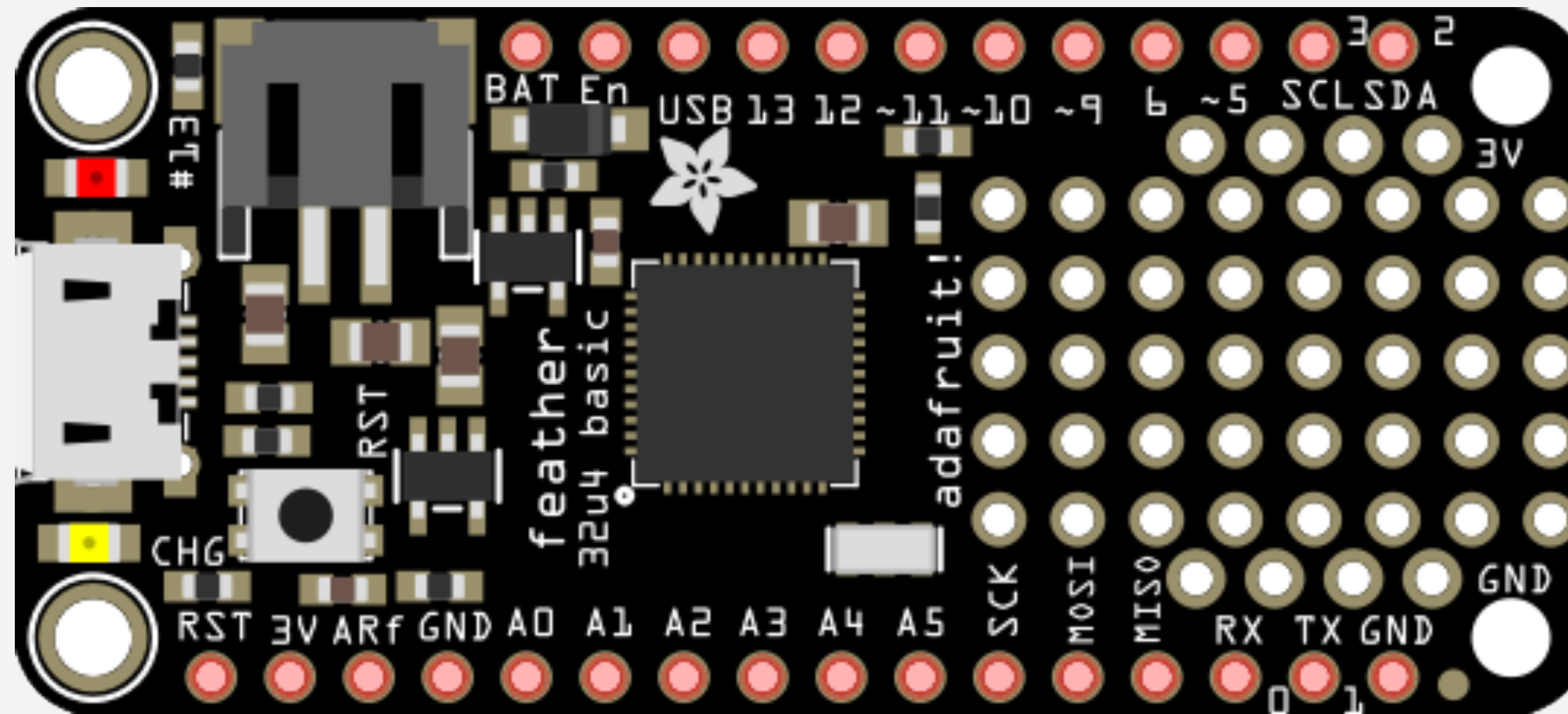


ARDUINO



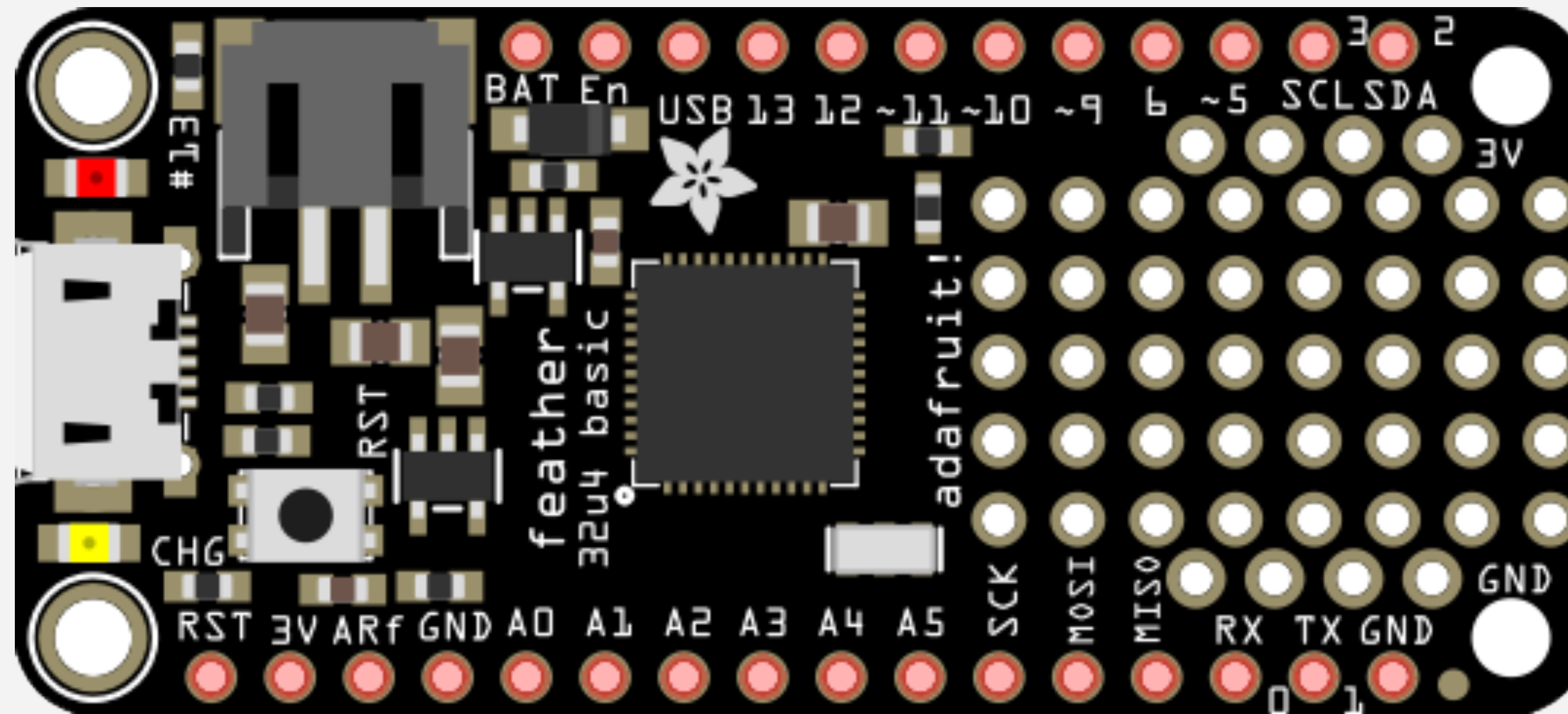
Arduino Uno

~~ARDUINO~~
FEATHER 32U4



Feather 32U4

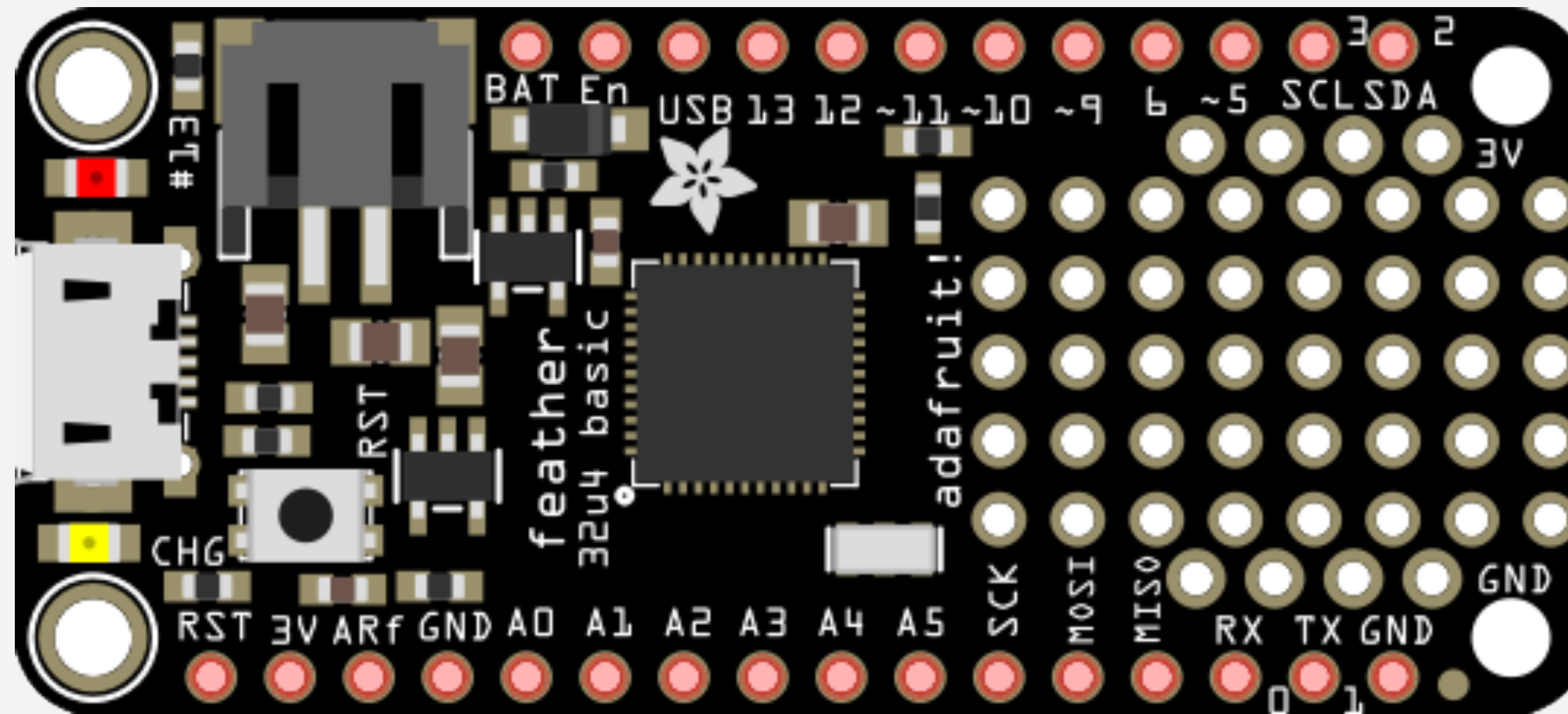
Pinout



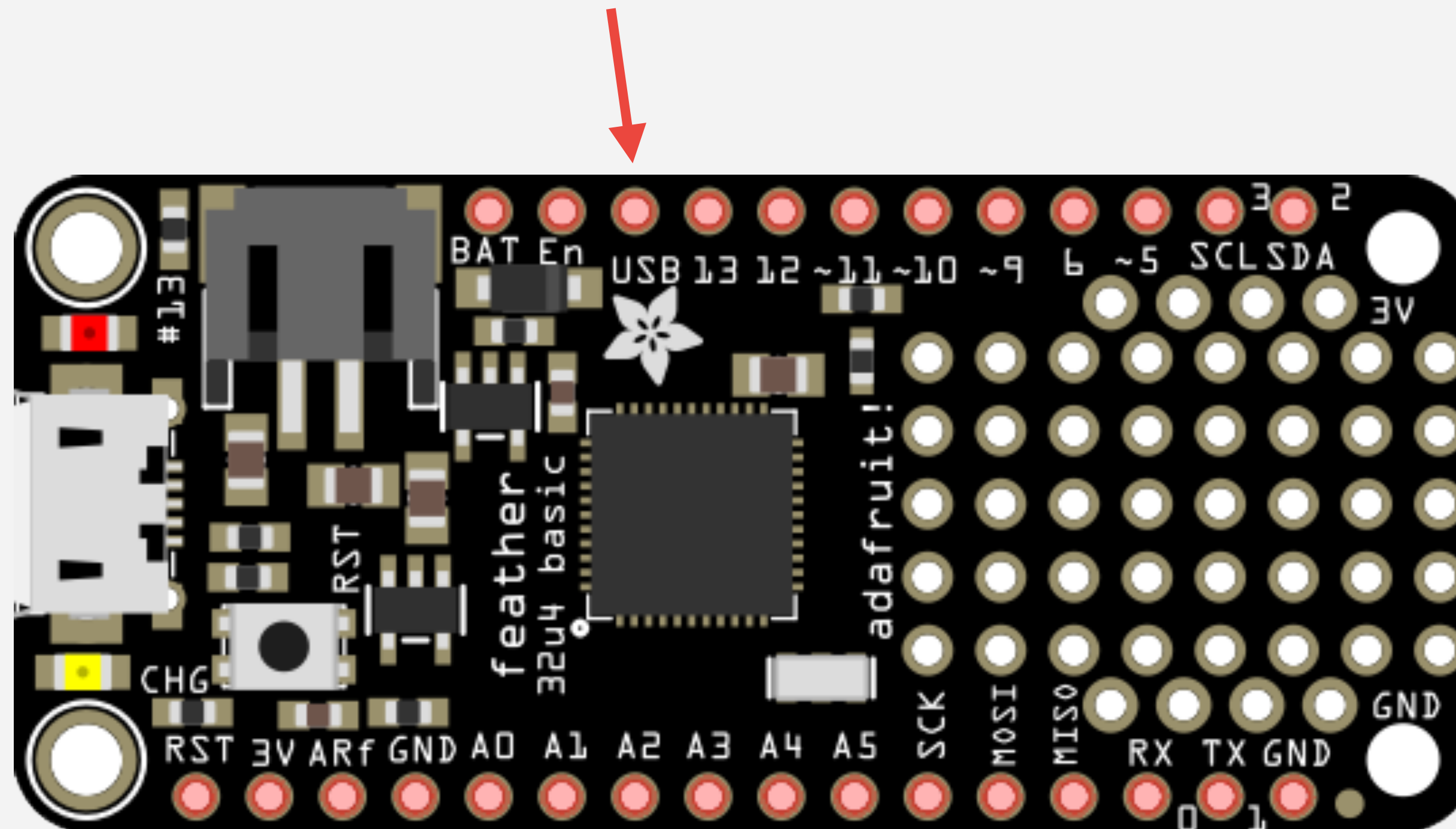
Feather 32U4



USB Port

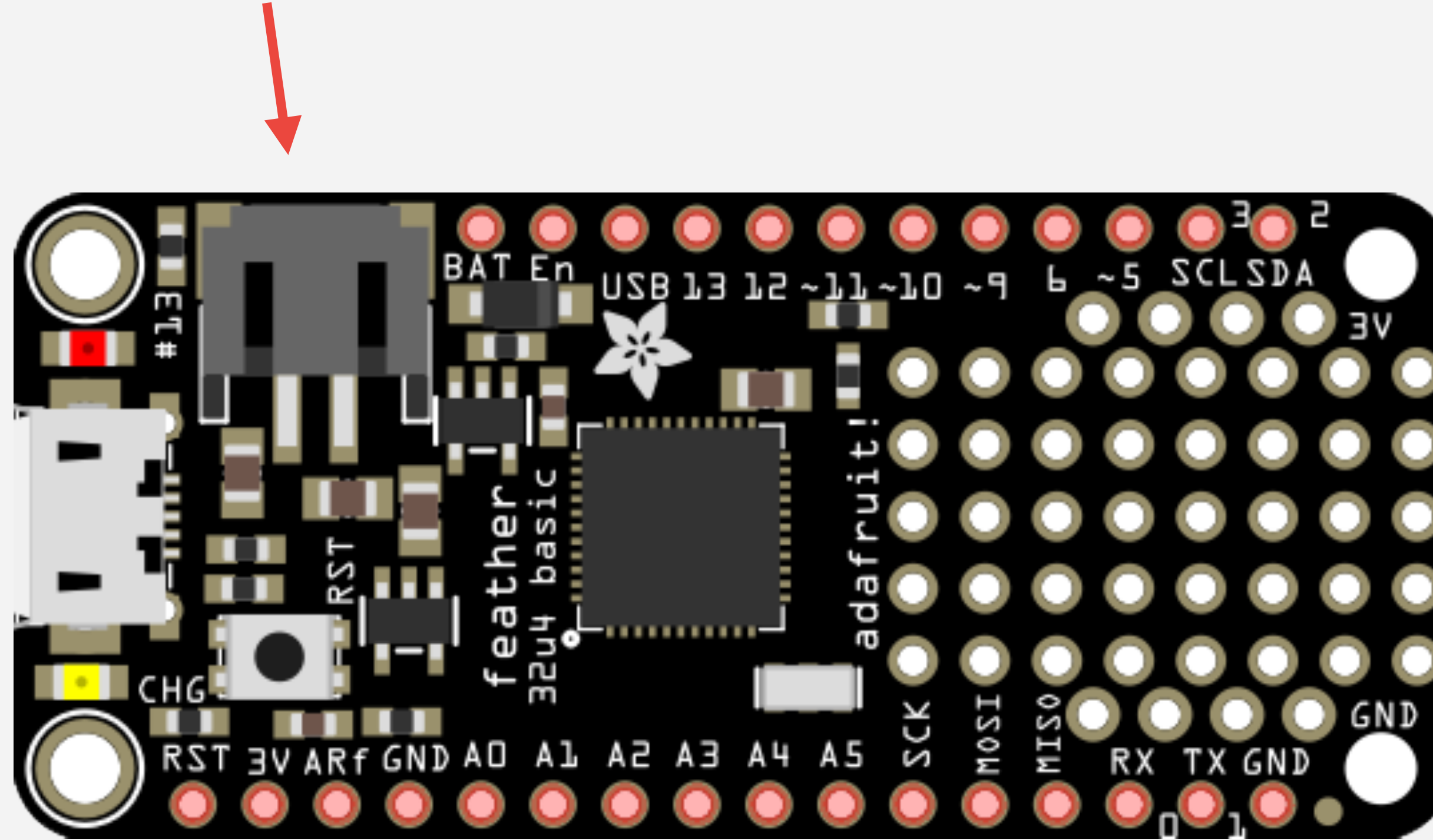


3.3V Power Output

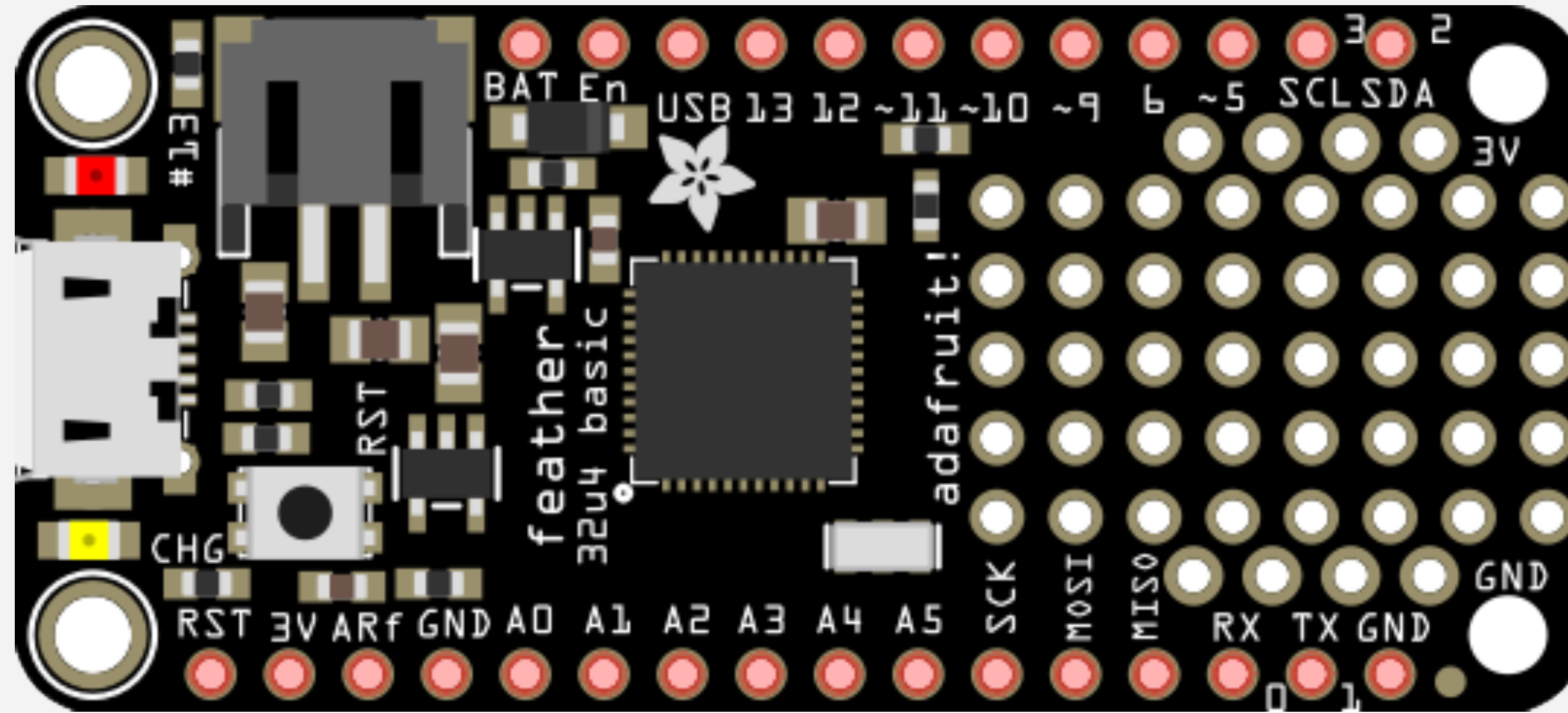


5V Power Output*

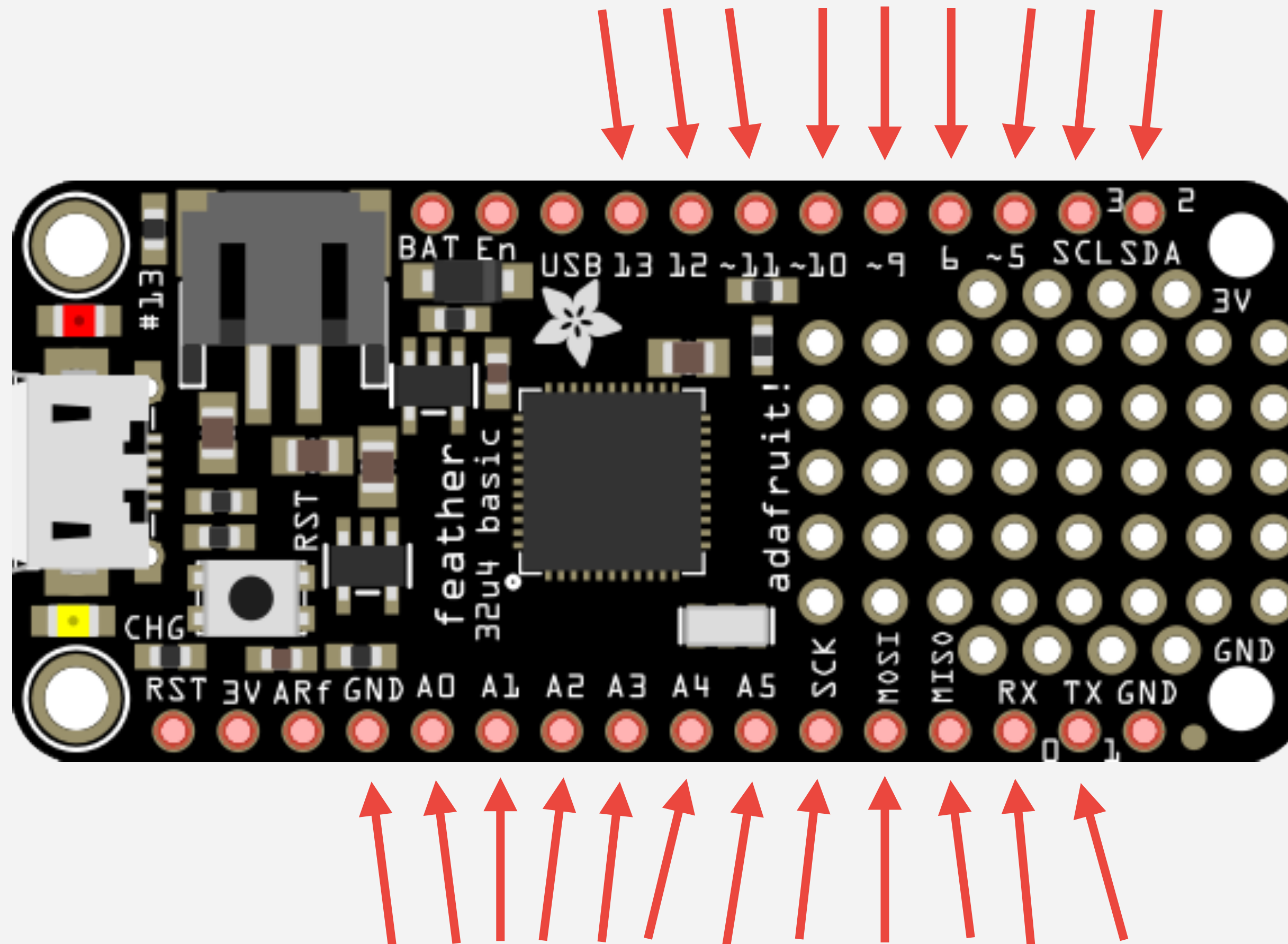
*When plugged in to USB



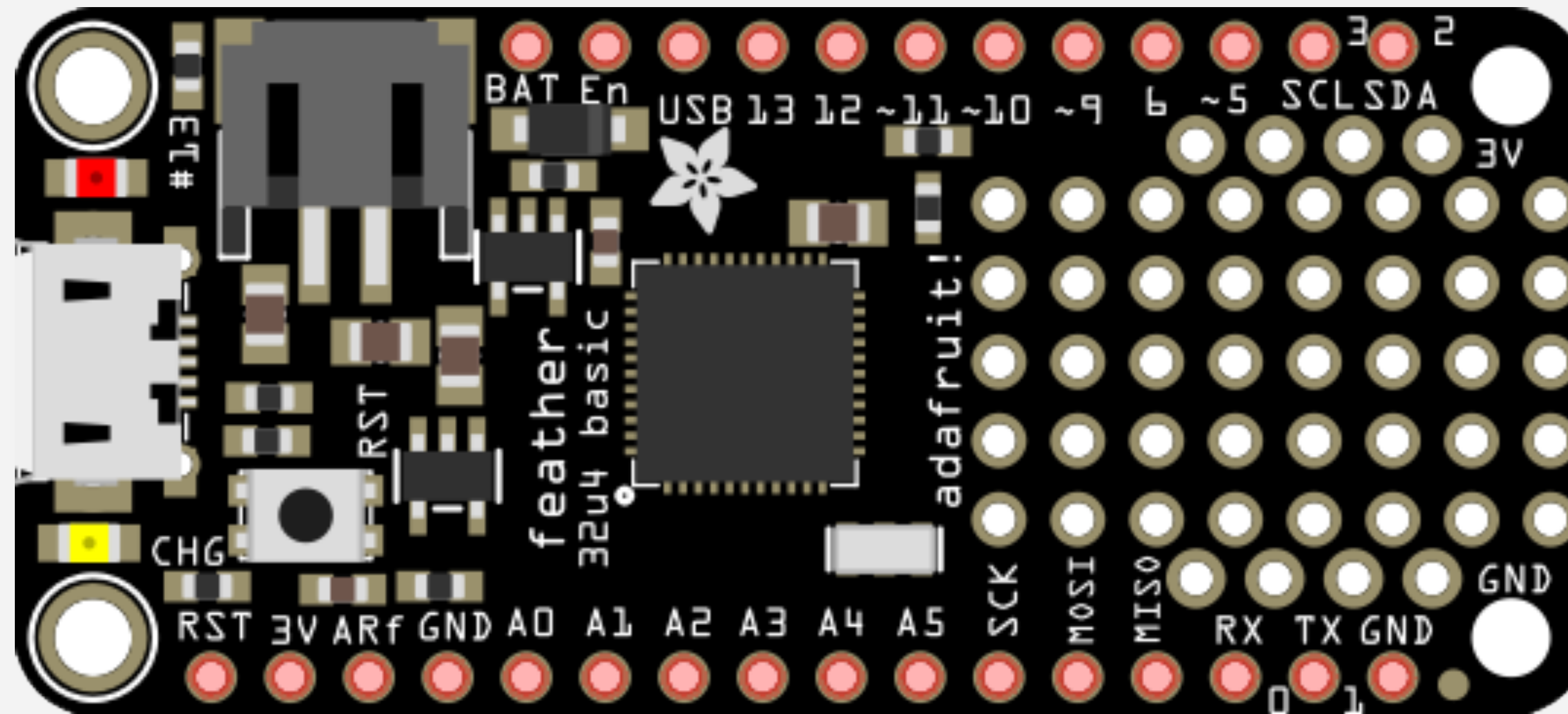
Battery Connector



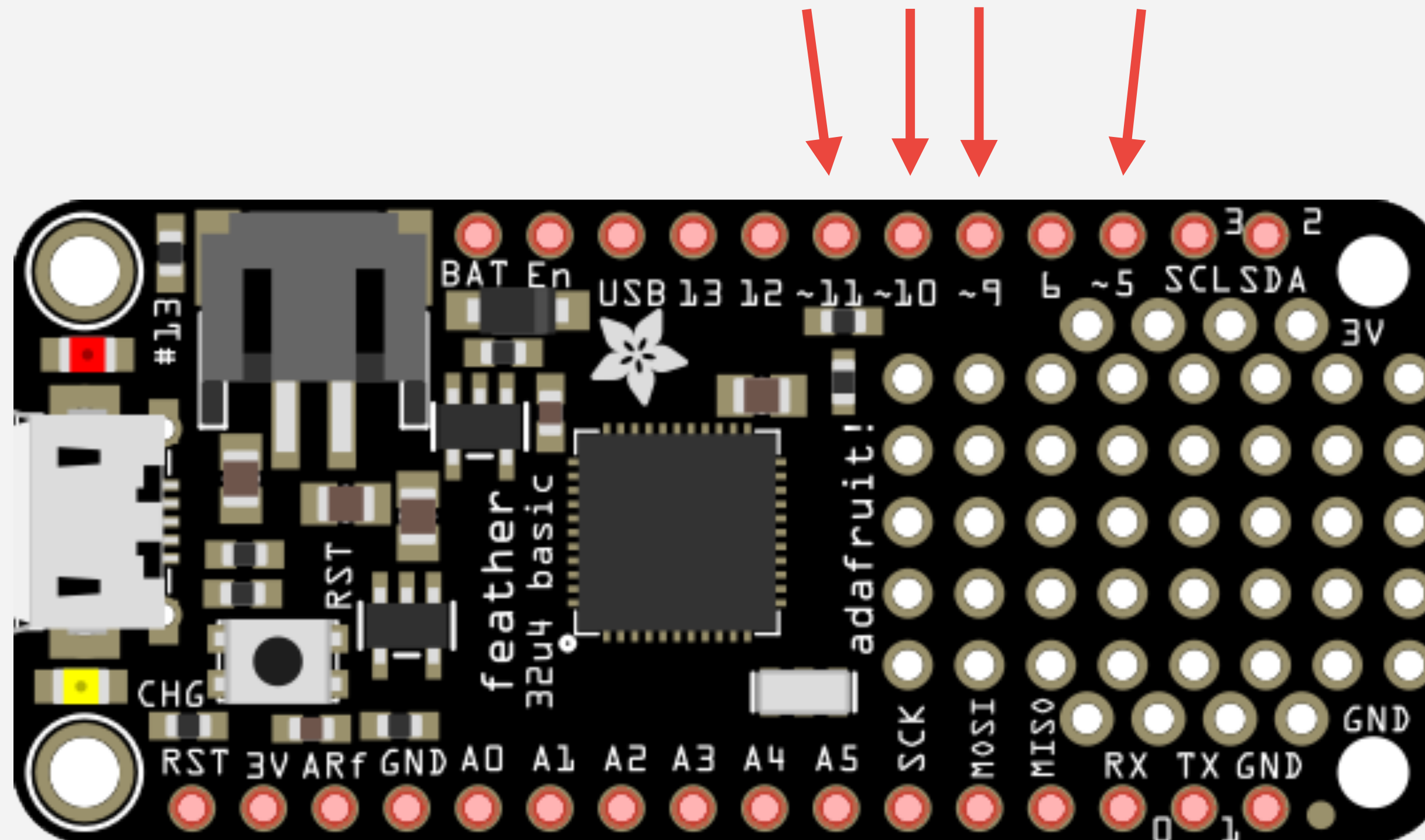
Ground



Digital Input and Output

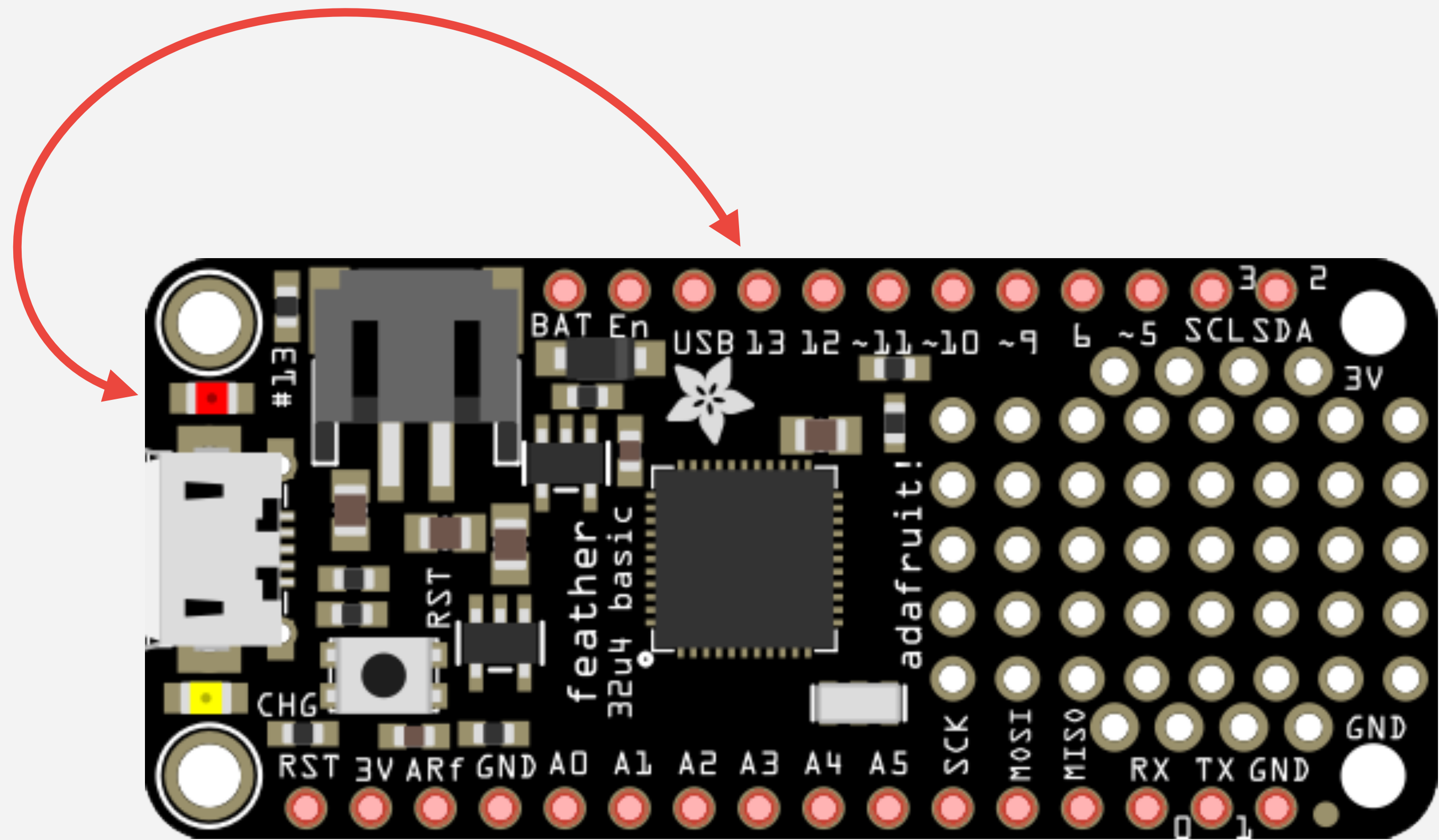


Analog Input



Analog Output*

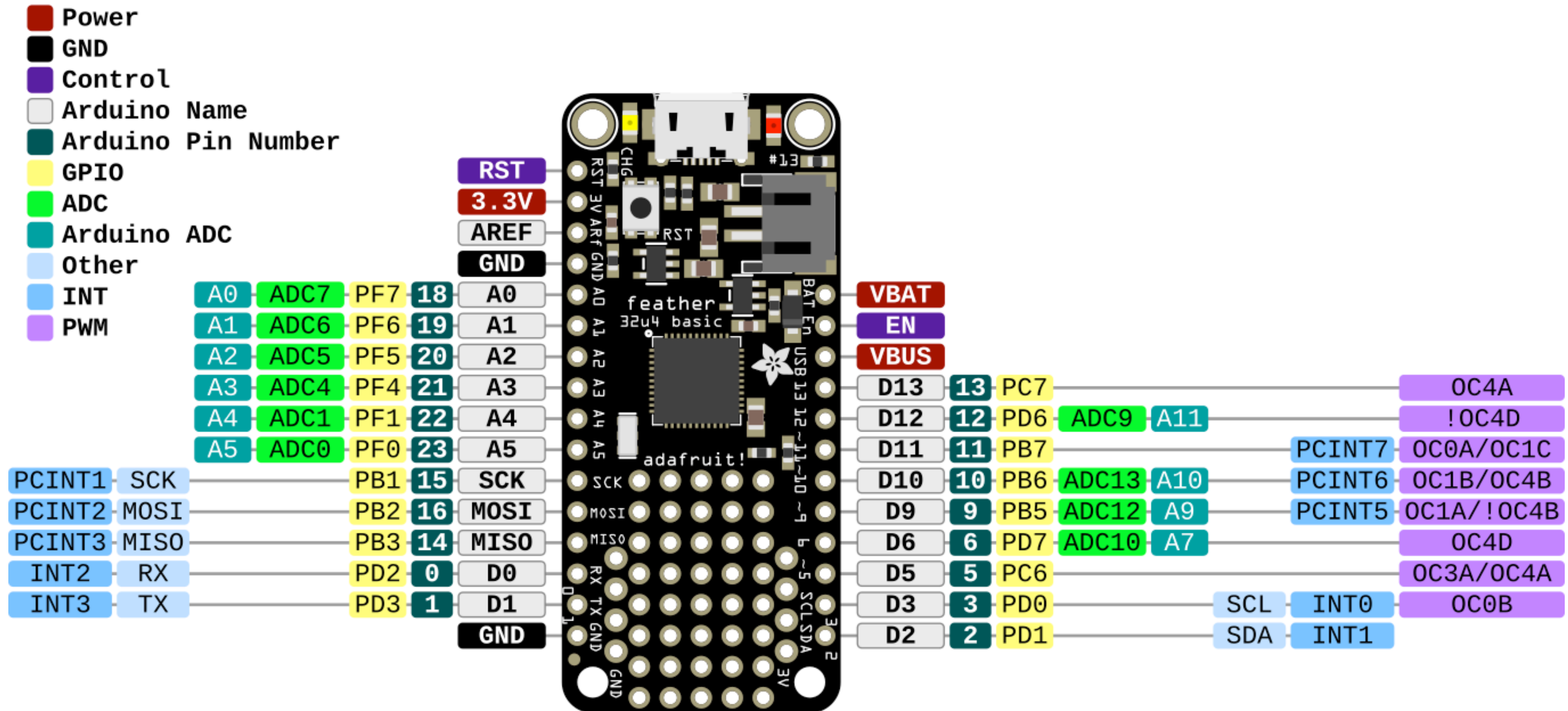
*Only on the pins with the ~ next to them



Pin 13 is connected to an LED

Adafruit Feather 32u4 Basic Proto

<https://www.adafruit.com/products/2771>



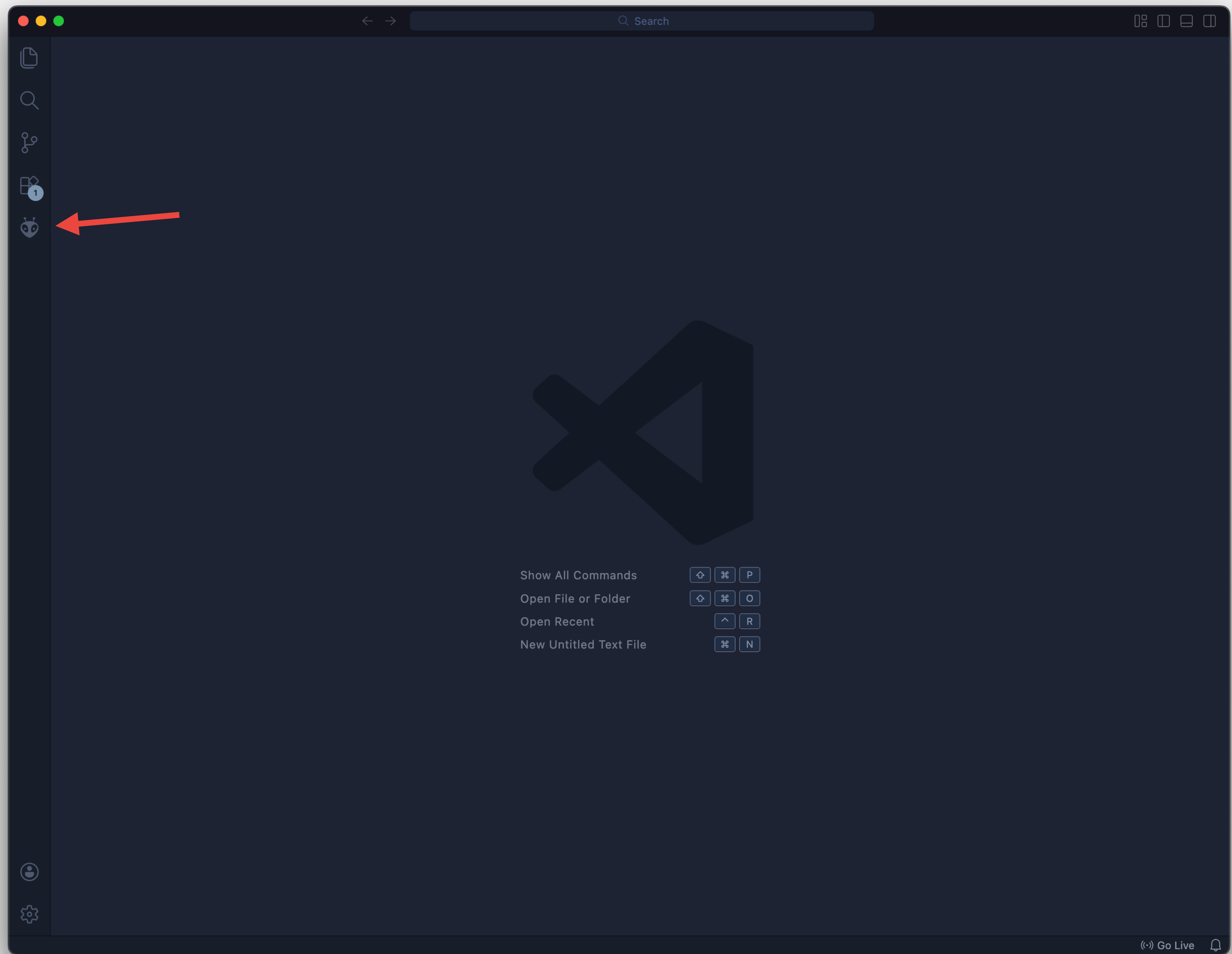
Tip!

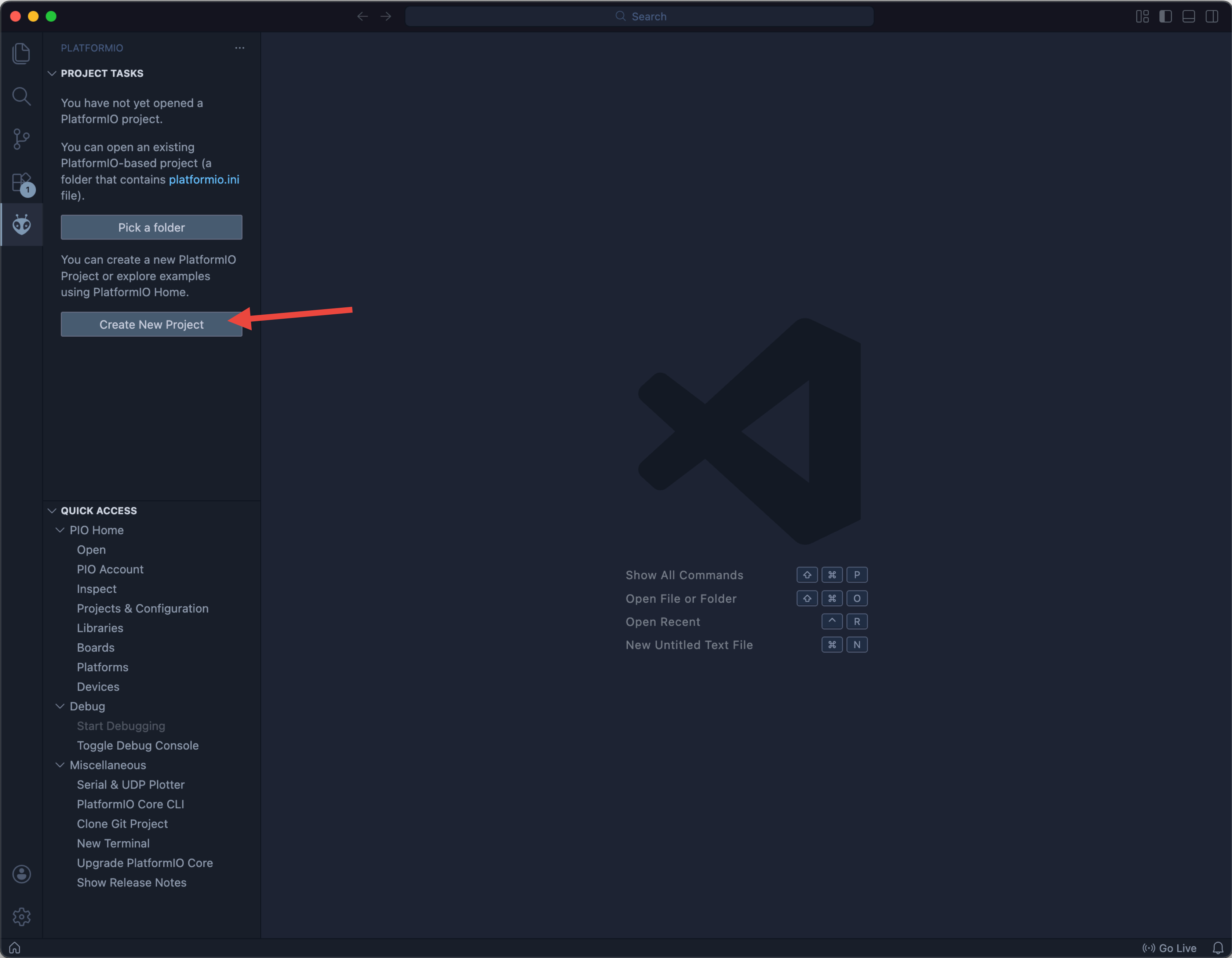
ALWAYS unplug the board
from USB before
connecting or disconnecting
things from the pins.

Setup

Install the ~~Arduino IDE~~
PlatformIO with
Feather Support

Create a new PlatformIO Project





PLATFORMIO

PROJECT TASKS

You have not yet opened a PlatformIO project.

You can open an existing PlatformIO-based project (a folder that contains `platformio.ini` file).

Pick a folder

You can create a new PlatformIO Project or explore examples using PlatformIO Home.

Create New Project

QUICK ACCESS

PIO Home

Open

PIO Account

Inspect

Projects & Configuration

Libraries

Boards

Platforms

Devices

Debug

Start Debugging

Toggle Debug Console

Miscellaneous

Serial & UDP Plotter

PlatformIO Core CLI

Clone Git Project

New Terminal

Upgrade PlatformIO Core

Show Release Notes

PIO Home

Home

Projects

Inspect

Libraries

Boards

Platforms

Devices

Search

Follow Us

Home

Welcome to PlatformIO

Core 6.1.18 · Home 3.4.4

Quick Access

+ New Project

Import Arduino Project

Open Project

Project Examples

Recent News

PlatformIO Labs · 3d

#LearnEmbedded "Understanding UART"

Explore the simplicity of #UART, a key communication protocol in embedded systems and computers. With just two wires, it enables

PlatformIO Labs · 6d

PlatformIO's 2025 Year in Review

★ 6M+ installs of the IDE for VSCode

🔧 20+ releases with long-term support

📦 Petabytes of traffic on our Trusted Package

PlatformIO Labs · 1w

#LearnEmbedded "Exploring Startup Implementations: Newlib (ARM)" by Phillip Johnston

Explore the program startup process and its

Recent Projects

Search project...

Name	Boards	Modified	Action
+ arduino/rough_year	Adafruit Feather M0 Express	7 months ago	Hide Open

Web · Open Source · Get Started · Docs · News · Community · Contact Us

If you enjoy using PlatformIO, please star our projects on GitHub!

☆ PlatformIO Core ☆

Go Live

PLATFORMIO

PROJECT TASKS

You have not yet opened a PlatformIO project.

You can open an existing PlatformIO-based project (a folder that contains `platformio.ini` file).

Pick a folder

You can create a new PlatformIO Project or explore examples using PlatformIO Home.

Create New Project

QUICK ACCESS

PIO Home

Open

PIO Account

Inspect

Projects & Configuration

Libraries

Boards

Platforms

Devices

Debug

Start Debugging

Toggle Debug Console

Miscellaneous

Serial & UDP Plotter

PlatformIO Core CLI

Clone Git Project

New Terminal

Upgrade PlatformIO Core

Show Release Notes

PIO Home

Home

Projects

Inspect

Libraries

Boards

Platforms

Devices

Welcome to PlatformIO

Project Wizard

This wizard allows you to **create new** PlatformIO project or **update existing**. In the last case, you need to uncheck "Use default location" and specify path to existing project.

Name:

Project name

Board:

Select a board (1709 available)

Framework:

Location:

☒ Use default location

Cancel

Finish

PlatformIO Labs

UART1

TX

RX

PlatformIO Labs

3d

#LearnEmbedded

Explore the simplicity of communication protocol and computers. With just two wires, it enables

Petabytes of traffic on our Trusted Package

Explore the program startup process and its

Recent Projects

Search project...

Name	Boards	Modified	Action
+ arduino/rough_year	Adafruit Feather M0 Express	7 months ago	Hide Open

Web

Open Source

Get Started

Docs

News

Community

Contact Us

If you enjoy using PlatformIO, please star our projects on GitHub!

PlatformIO Core

Go Live

PLATFORMIO

PROJECT TASKS

You have not yet opened a PlatformIO project.

You can open an existing PlatformIO-based project (a folder that contains `platformio.ini` file).

Pick a folder

You can create a new PlatformIO Project or explore examples using PlatformIO Home.

Create New Project

QUICK ACCESS

PIO Home

Open

PIO Account

Inspect

Projects & Configuration

Libraries

Boards

Platforms

Devices

Debug

Start Debugging

Toggle Debug Console

Miscellaneous

Serial & UDP Plotter

PlatformIO Core CLI

Clone Git Project

New Terminal

Upgrade PlatformIO Core

Show Release Notes

PIO Home

Home

Projects

Inspect

Libraries

Boards

Platforms

Devices

Welcome to PlatformIO

Project Wizard

This wizard allows you to **create new** PlatformIO project or **update existing**. In the last case, you need to uncheck "Use default location" and specify path to existing project.

Name:

Blink

Board:

Adafruit Feather 32u4

Framework:

Arduino

Location:

☒ Use default location ?

Cancel

Finish

PlatformIO Labs

3d

#LearnEmbedded

Explore the simplicity of communication protocol and computers. With just two wires, it enables

Petabytes of traffic on our Trusted Package

Explore the program startup process and its

Recent Projects

Search project...

Name	Boards	Modified	Action
+ arduino/rough_year	Adafruit Feather M0 Express	7 months ago	Hide Open

Web

Open Source

Get Started

Docs

News

Community

Contact Us

If you enjoy using PlatformIO, please star our projects on GitHub!

☆

 PlatformIO Core

☆

Go Live

PLATFORMIO

PROJECT TASKS

You have not yet opened a PlatformIO project.

You can open an existing PlatformIO-based project (a folder that contains `platformio.ini` file).

Pick a folder

You can create a new PlatformIO Project or explore examples using PlatformIO Home.

Create New Project

QUICK ACCESS

PIO Home

Open

PIO Account

Inspect

Projects & Configuration

Libraries

Boards

Platforms

Devices

Debug

Start Debugging

Toggle Debug Console

Miscellaneous

Serial & UDP Plotter

PlatformIO Core CLI

Clone Git Project

New Terminal

Upgrade PlatformIO Core

Show Release Notes

PIO Home

Home

Projects

Inspect

Libraries

Boards

Platforms

Devices

Welcome to PlatformIO

Project Wizard

This wizard allows you to **create new** PlatformIO project or **update existing**. In the last case, you need to uncheck "Use default location" and specify path to existing project.

Name: Blink

Board:

Adafruit Feather 32u4

Framework:

Arduino

Location: ☒ Use default location ?

Cancel

Finish

PlatformIO Labs • 3d

#LearnEmbedded "Un

Explore the simplicity of communication protocol and computers. With just two wires, it enables

Petabytes of traffic on our Trusted Package

Explore the program startup process and its

Recent Projects

Search project...

Name	Boards	Modified	Action
+ arduino/rough_year	Adafruit Feather M0 Express	7 months ago	Hide Open

Web

 ·

Open Source

 ·

Get Started

 ·

Docs

 ·

News

 ·

Community

 ·

Contact Us

If you enjoy using PlatformIO, please star our projects on GitHub!

☆

PlatformIO Core

☆

Go Live

PLATFORMIO

PROJECT TASKS

You have not yet opened a PlatformIO project.

You can open an existing PlatformIO-based project (a folder that contains `platformio.ini` file).

Pick a folder

You can create a new PlatformIO Project or explore examples using PlatformIO Home.

Create New Project

QUICK ACCESS

PIO Home

Open

PIO Account

Inspect

Projects & Configuration

Libraries

Boards

Platforms

Devices

Debug

Start Debugging

Toggle Debug Console

Miscellaneous

Serial & UDP Plotter

PlatformIO Core CLI

Clone Git Project

New Terminal

Upgrade PlatformIO Core

Show Release Notes

PIO Home

Home

Projects

Inspect

Libraries

Boards

Platforms

Devices

Welcome to PlatformIO

UART1

TX

RX

PlatformIO Labs • 3d

#LearnEmbedded "Ur

Explore the simplicity of

communication protocol

and computers. With just two wires, it enables

Petabytes of traffic on our Trusted Package

Explore the program startup process and its

Recent Projects

Search project...

Name	Boards	Modified	Action
+ arduino/rough_year	Adafruit Feather M0 Express	7 months ago	Hide Open

Web · Open Source · Get Started · Docs · News · Community · Contact Us

If you enjoy using PlatformIO, please star our projects on GitHub!

☆ PlatformIO Core ☆

Project Wizard

This wizard allows you to **create new** PlatformIO project or **update existing**. In the last case, you need to uncheck "Use default location" and specify path to existing project.

Name:

Blink

Board:

Adafruit Feather 32u4

Framework:

Arduino

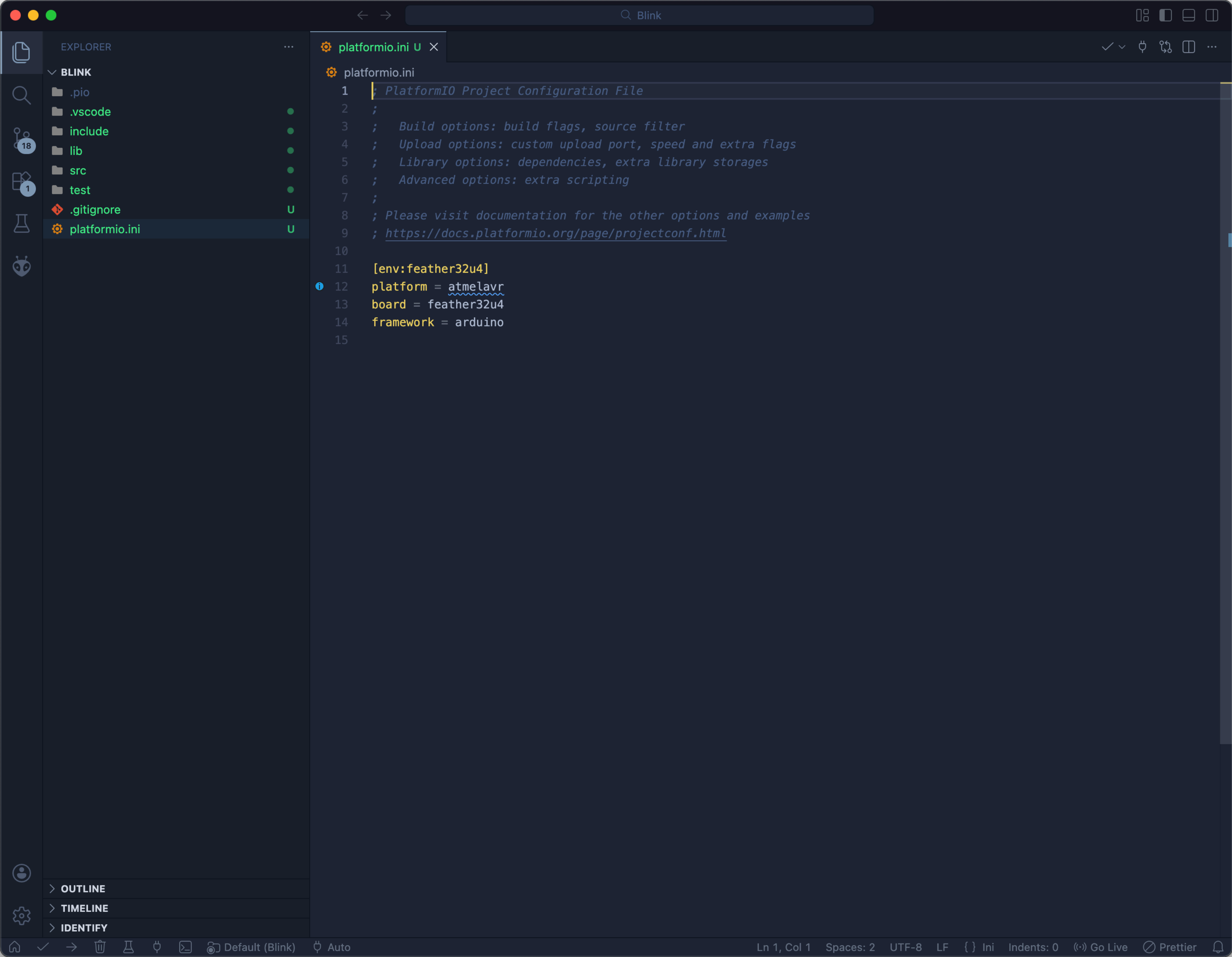
Location:

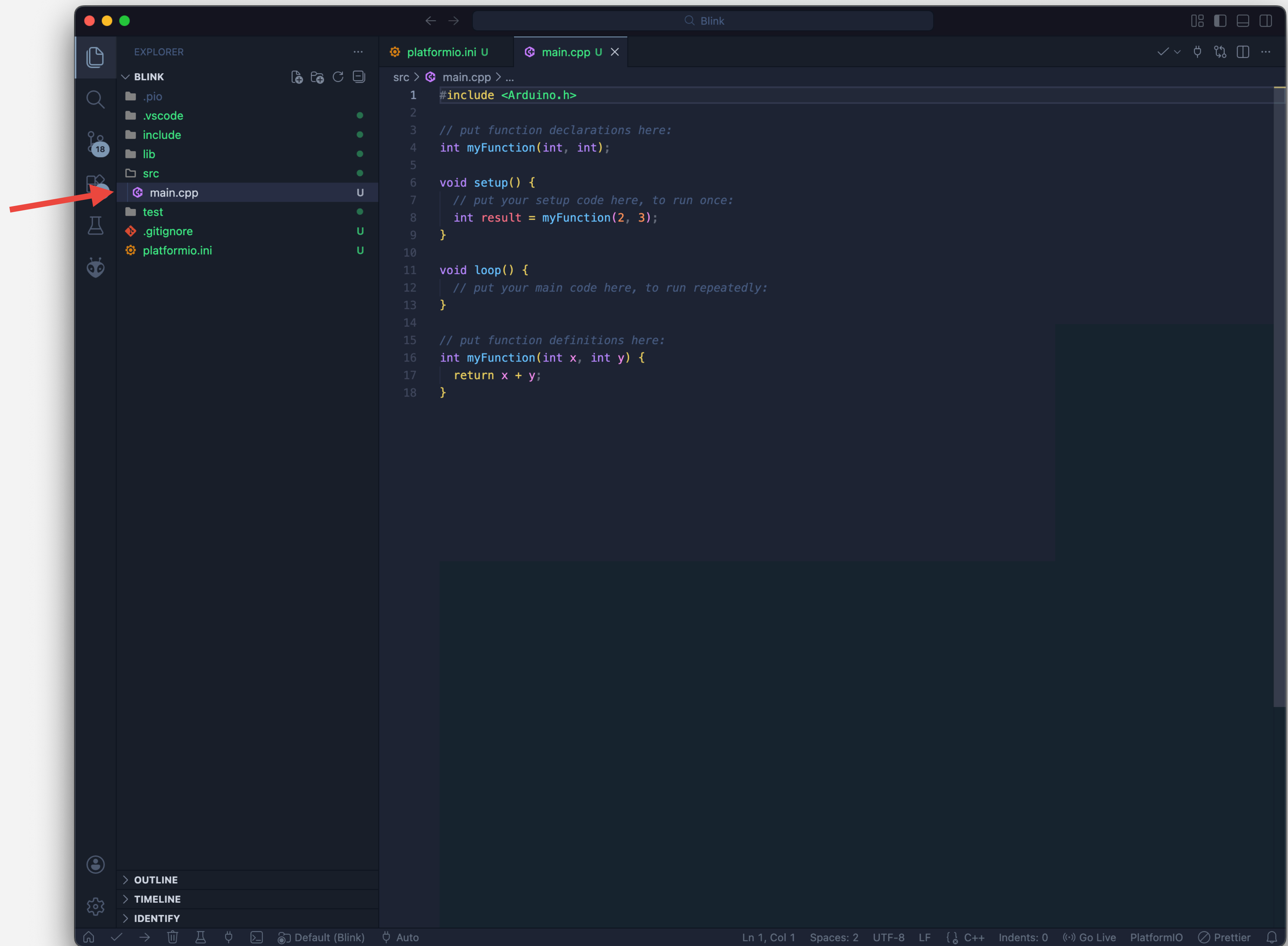
☒ Use default location ?

Cancel

Finish

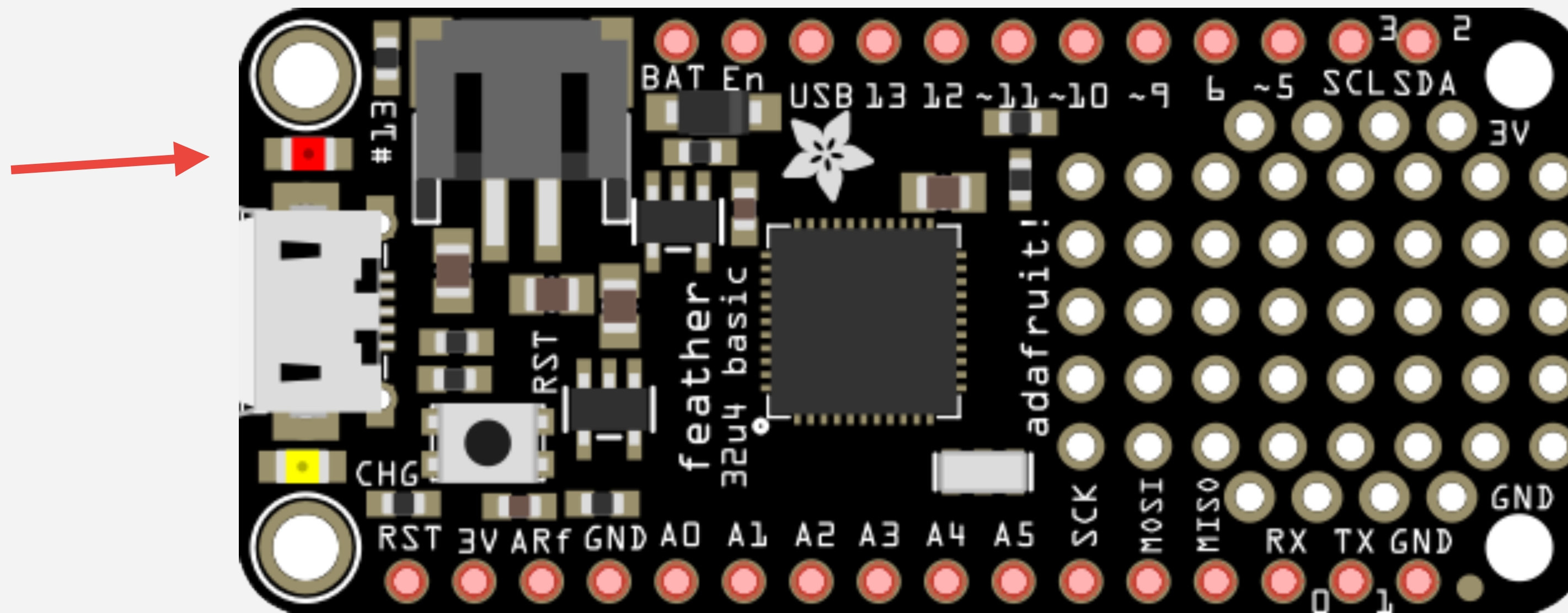
Go Live



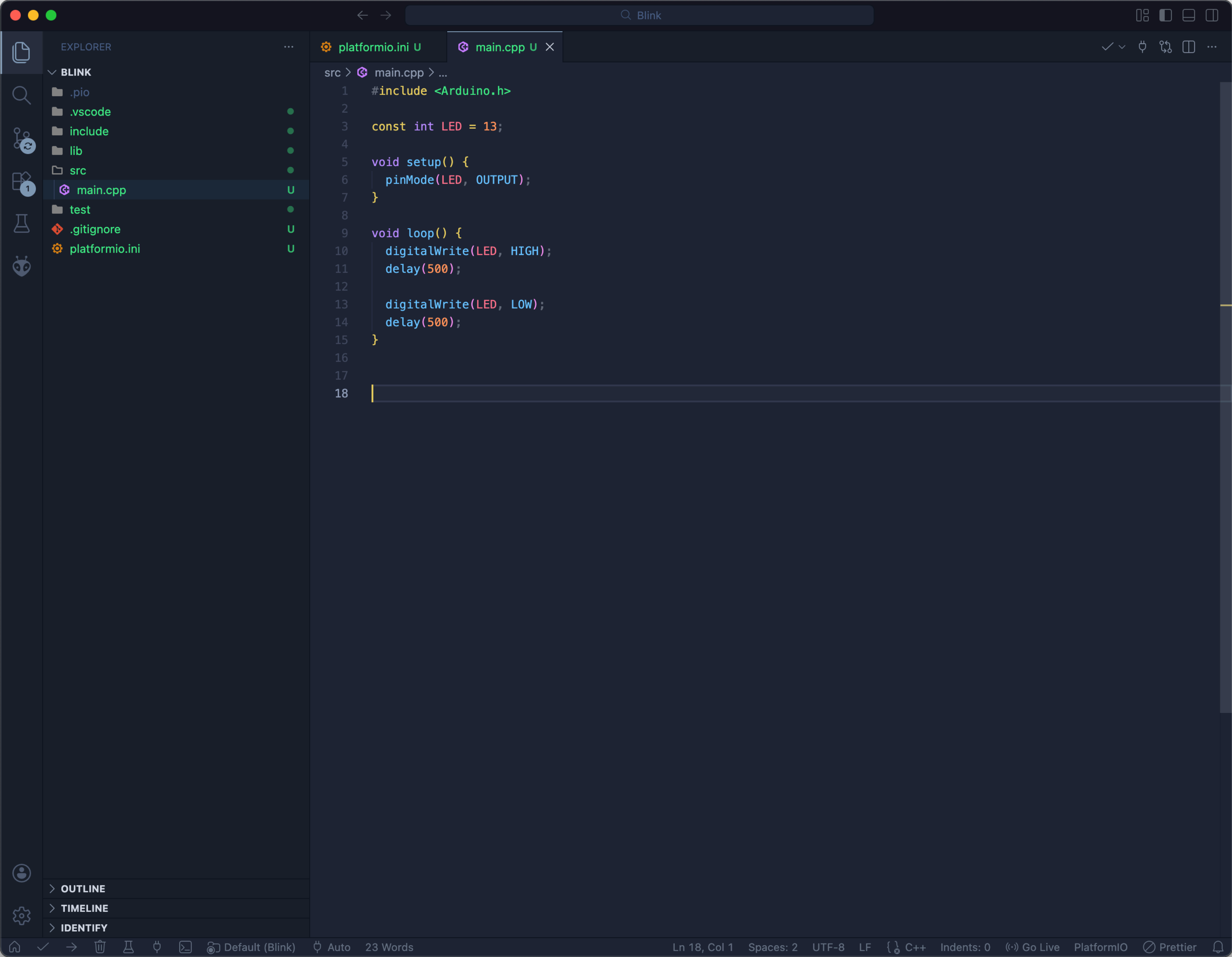


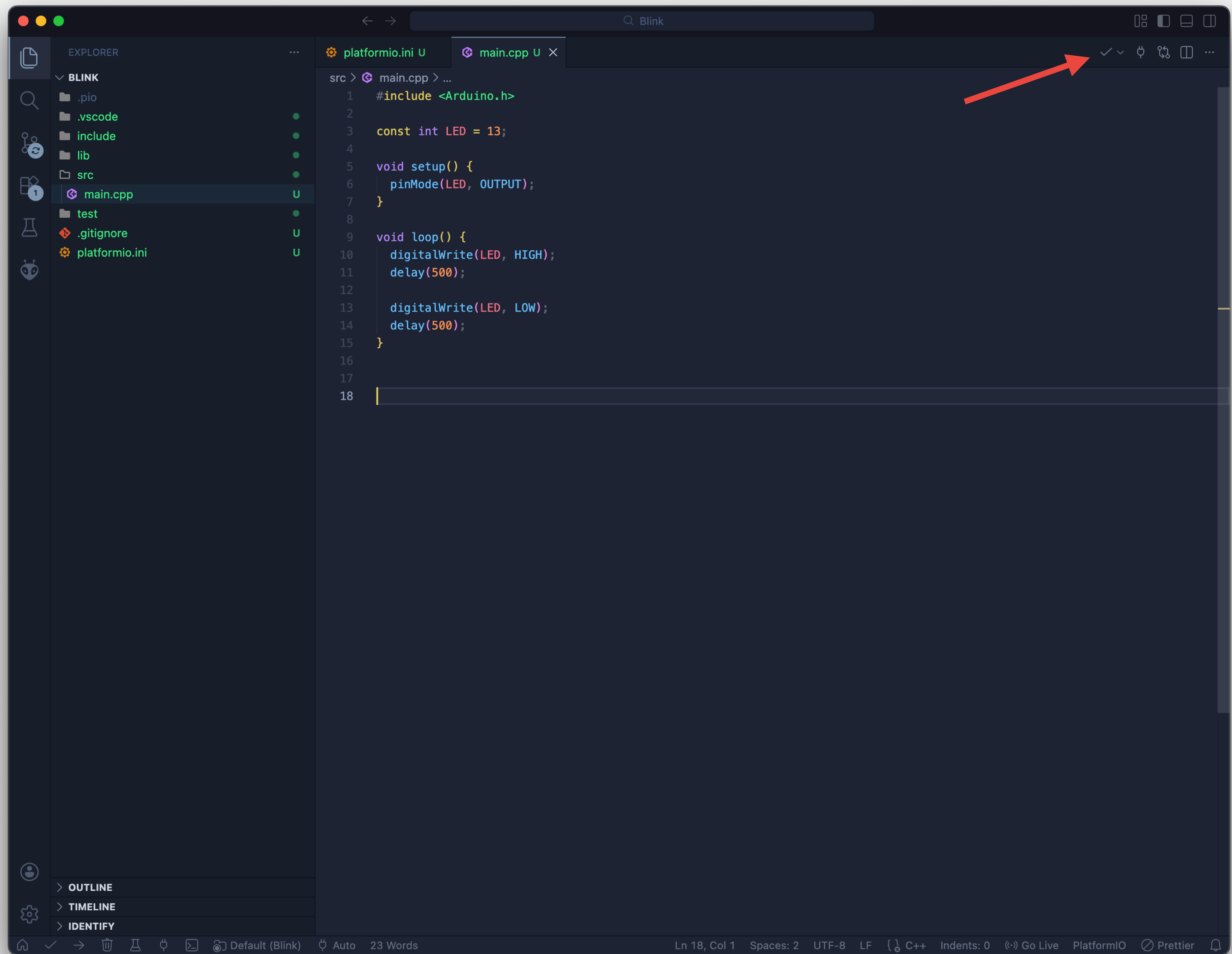
HELLO LED!

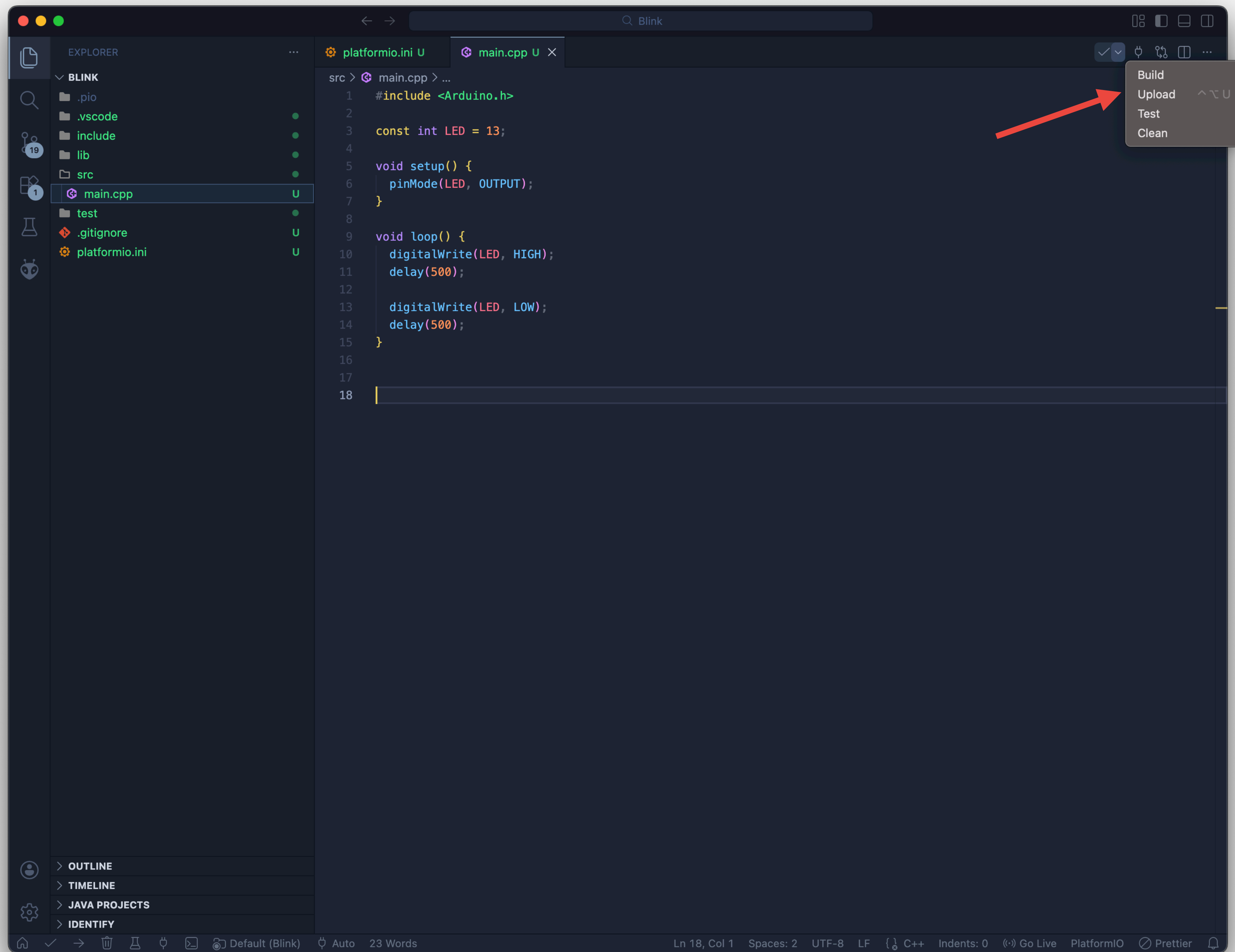
Blink the built-in LED



Built-in LED







18

1

EXPLOSER

BLINK

.pio

.vscode

include

lib

src

main.cpp

test

.gitignore

platformio.ini

PlatformIO: Upload

src > main.cpp > LED

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

#include <Arduino.h>

const int LED = 13;

void setup() {

pinMode(LED, OUTPUT);

}

void loop() {

digitalWrite(LED, HIGH);

delay(500);

digitalWrite(LED, LOW);

delay(500);

}

TERMINAL 1

TERMINAL

* Executing task: platformio run

* The terminal process "platformio 'run'" terminated with exit code: 1.

* Terminal will be reused by tasks, press any key to close it.

* Executing task: platformio run

Processing feather32u4 (platform: atmelavr; board: feather32u4; framework: arduino)

Verbose mode can be enabled via '-v, --verbose' option

CONFIGURATION: <https://docs.platformio.org/page/boards/atmelavr/feather32u4.html>

PLATFORM: Atmel AVR (5.1.0) > Adafruit Feather 32u4

HARDWARE: ATMEGA32U4 8MHz, 2.50 KB RAM, 28KB Flash

DEBUG: Current (simavr) External (simavr)

PACKAGES:

- framework-arduino-avr @ 5.2.0

- toolchain-atmelavr @ 1.70300.191015 (7.3.0)

LDF: Library Dependency Finder -> <https://bit.ly/configure-pio-ldf>

LDF Modes: Finder ~ chain, Compatibility ~ soft

Found 5 compatible libraries

Scanning dependencies...

No dependencies

Building in release mode

Compiling .pio/build/feather32u4/src/main.cpp.o

Archiving .pio/build/feather32u4/libFrameworkArduino.a

Indexing .pio/build/feather32u4/libFrameworkArduino.a

Linking .pio/build/feather32u4/firmware.elf

Checking size .pio/build/feather32u4/firmware.elf

Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"

RAM: [] 5.8% (used 149 bytes from 2560 bytes)

Flash: [] 13.5% (used 3862 bytes from 28672 bytes)

Building .pio/build/feather32u4/firmware.hex

= [SUCCESS] Took 0.73 seconds

* Terminal will be reused by tasks, press any key to close it.

PROBLEMS

OUTPUT

1

Default (Blink)

Auto

23 Words

Ln 3, Col 19

Spaces: 2

UTF-8

LF

C++

Indents: 0

Go Live

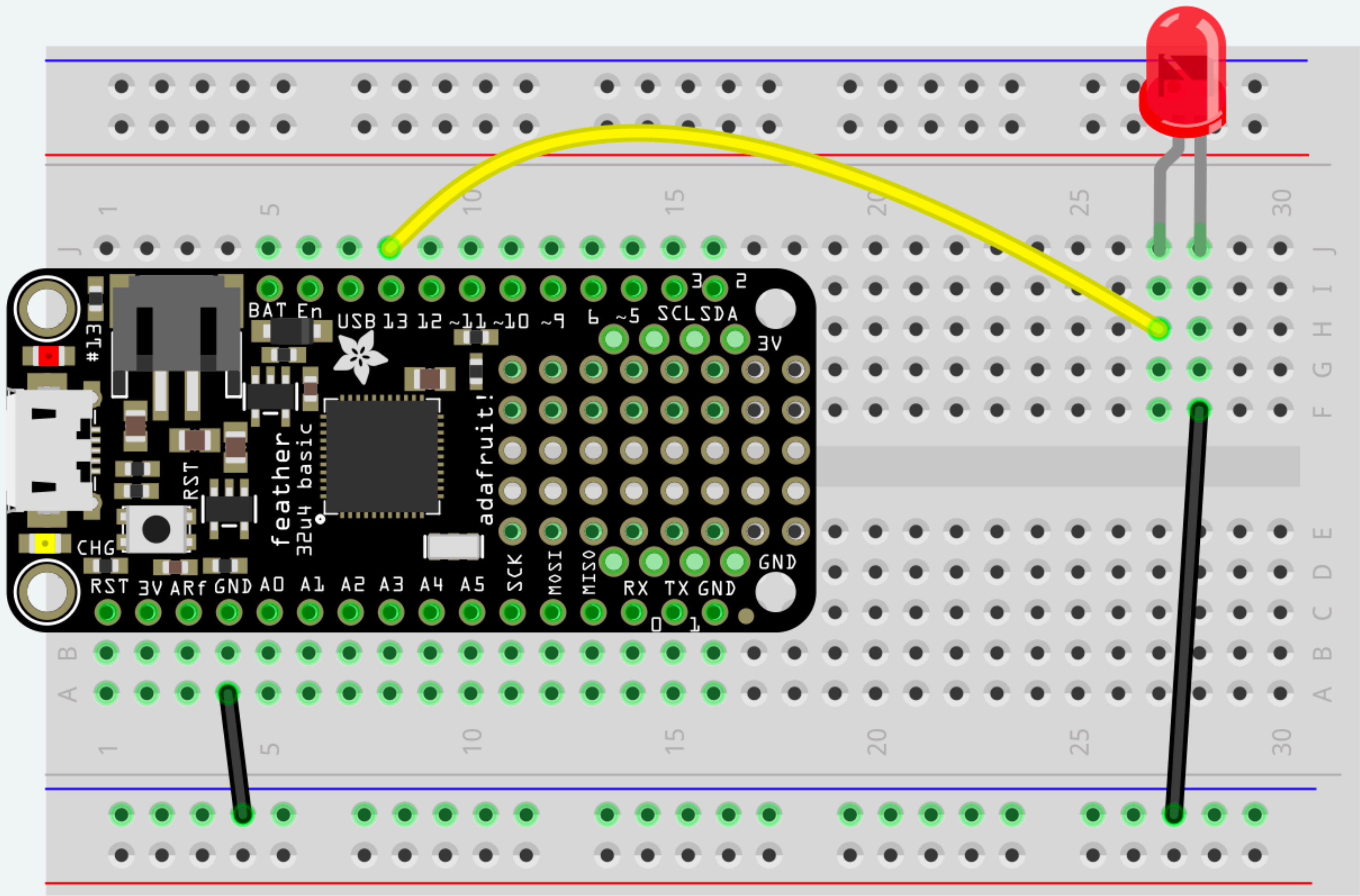
PlatformIO

Prettier

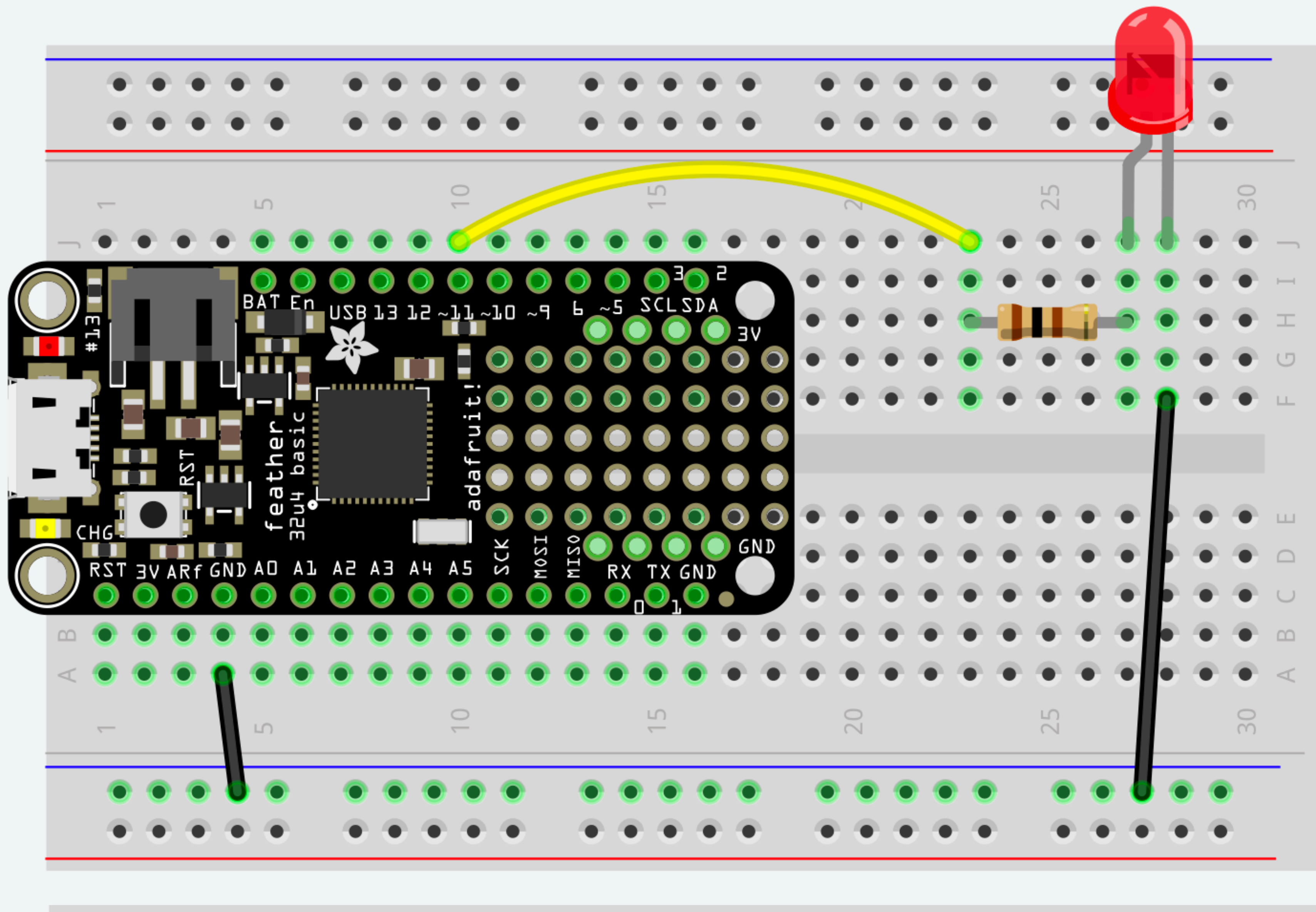
Try changing the blink frequency,
then try creating a pattern.

Remember that the `delay()`
function is what makes the code
pause.

Blink an external LED on Pin 13



**Blink an external LED
on other pins**



* Resistor is 100Ω (Brown Black Brown)

🔍

🔗

📁

📄

🔧

🤖

EXPLORER

BLINK

.pio

.vscode

include

lib

src

main.cpp

test

.gitignore

platformio.ini

platformio.ini

main.cpp

src > main.cpp > ...

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

#include <Arduino.h>

const int LED = 9;

void setup() {

pinMode(LED, OUTPUT);

}

void loop() {

digitalWrite(LED, HIGH);

delay(500);

digitalWrite(LED, LOW);

delay(500);

}

TERMINAL 1

TERMINAL

* Executing task: platformio run --target upload

er function 'void USBDevice::attach()':

/Users/chandler/.platformio/packages/framework-arduino-avr/cor

es/arduino/USBCore.cpp:835:16:

warning: statement has no effect [-Wunused-value]

TX_RX_LED_INIT;

Compiling .pio/build/feather32u4/FrameworkArduino/wiring_digital.c.o

Compiling .pio/build/feather32u4/FrameworkArduino/wiring_pulse.S.o

Compiling .pio/build/feather32u4/FrameworkArduino/wiring_pulse.c.o

PROBLEMS

Filter (e.g. text, **/*.ts, !**/node_modu... 🔍)

platformio.ini

"atmelavr":... cSpell [Ln 12, Col 12]

OUTPUT

Filter (e.g. text, !excludeText, text1,tex...)

Activating task providers all

Activating task providers java

🔧 2

🏠

✓

➡

🗑

🧪

🔌

📧

📄

Default (Blink)

🔌 Auto

23 Words

🔄 Building...

Ln 18, Col 1

Spaces: 2

UTF-8

LF

{ } C++

Indents: 0

🔄 Go Live

PlatformIO

🔄 Prettier

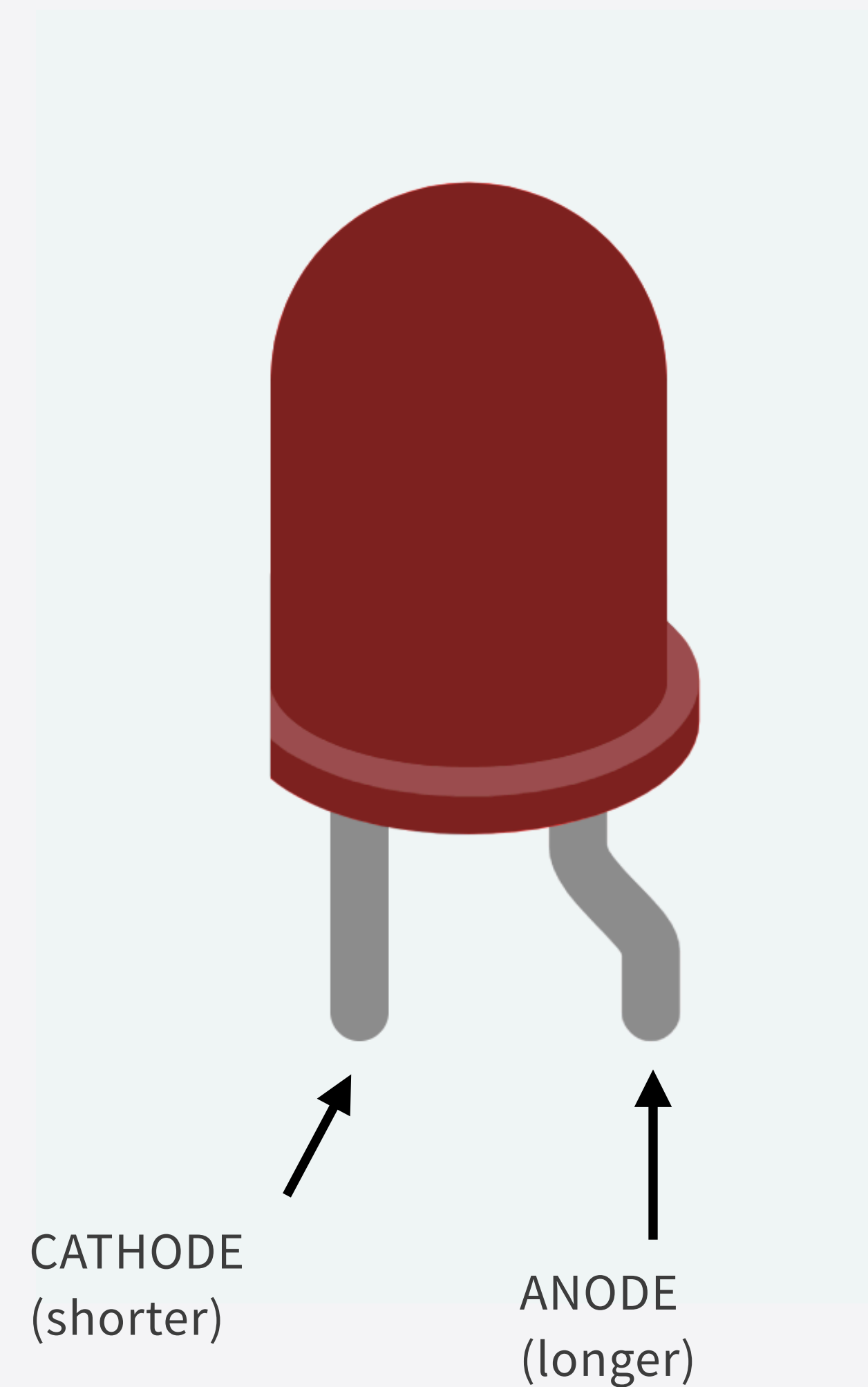
🔔

Try connecting more LEDs
to other pins.

What patterns can you create?

What limits/complications are caused
by using the `delay()` function?

An LED has two legs, a longer one called an anode and a shorter one called a cathode



Short Circuit!

This can happen by accidentally connecting the positive and negative terminals of a battery, or, more often, but some loose wires coming out and touching.

Make sure your wires are always trimmed and minimize the amount of metal sticking out.

